Logistic Regression

- Classification

로지스틱 회귀 - Classification

- Binary
- 스팸 메일 탐지 : Spam or Ham
- Facebook feed : show or hide
- 신용카드 사기 거래 탐지 :

  정상 거래 / 사기 거래

로지스틱 회귀 - Classification

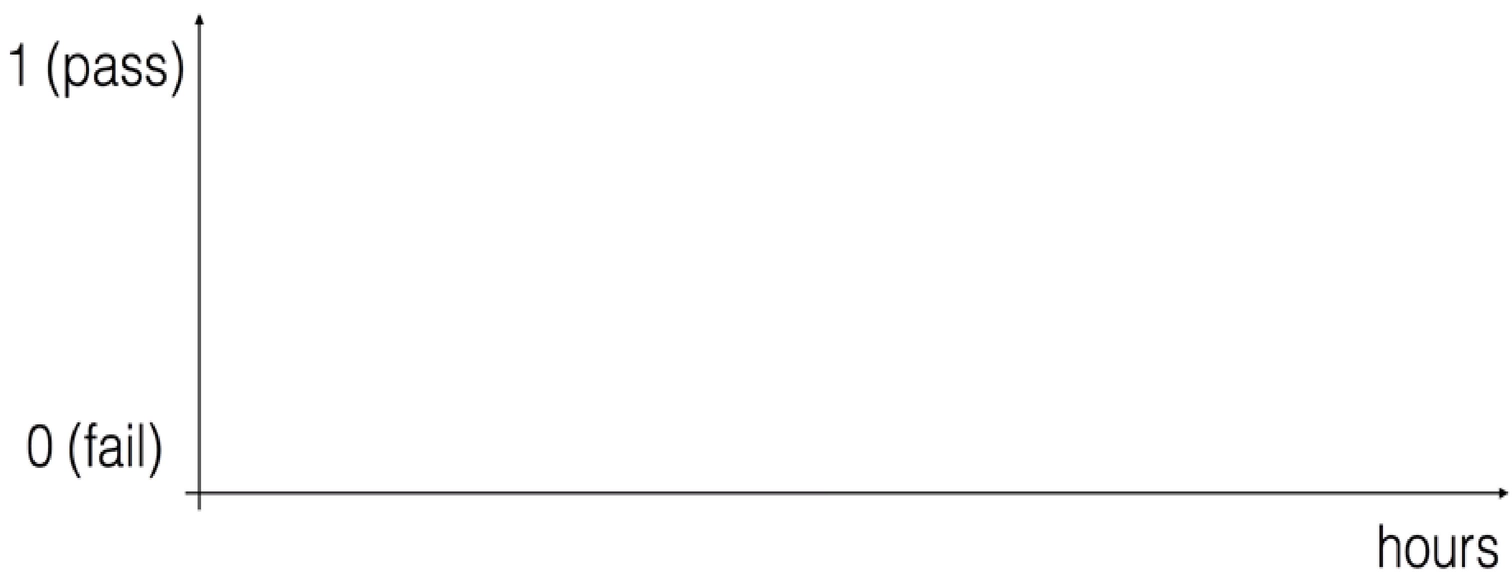
0 / 1 encoding

- 스팸 메일 탐지 : Spam(1) or Ham(0)
- Facebook feed : show(1) or hide(0)
- 신용카드 사기 거래 탐지 :

  정상적 거래(0) / 사기 거래(1)

로지스틱 회귀 - Classification

Pass(1)/Fail(0) based on study hours



로지스틱 회귀 - Classification

Linear regression

- We know Y is 0 or 1

  $H(x) = Wx + b$

- Hypothesis can give values large than 1 or less than 0

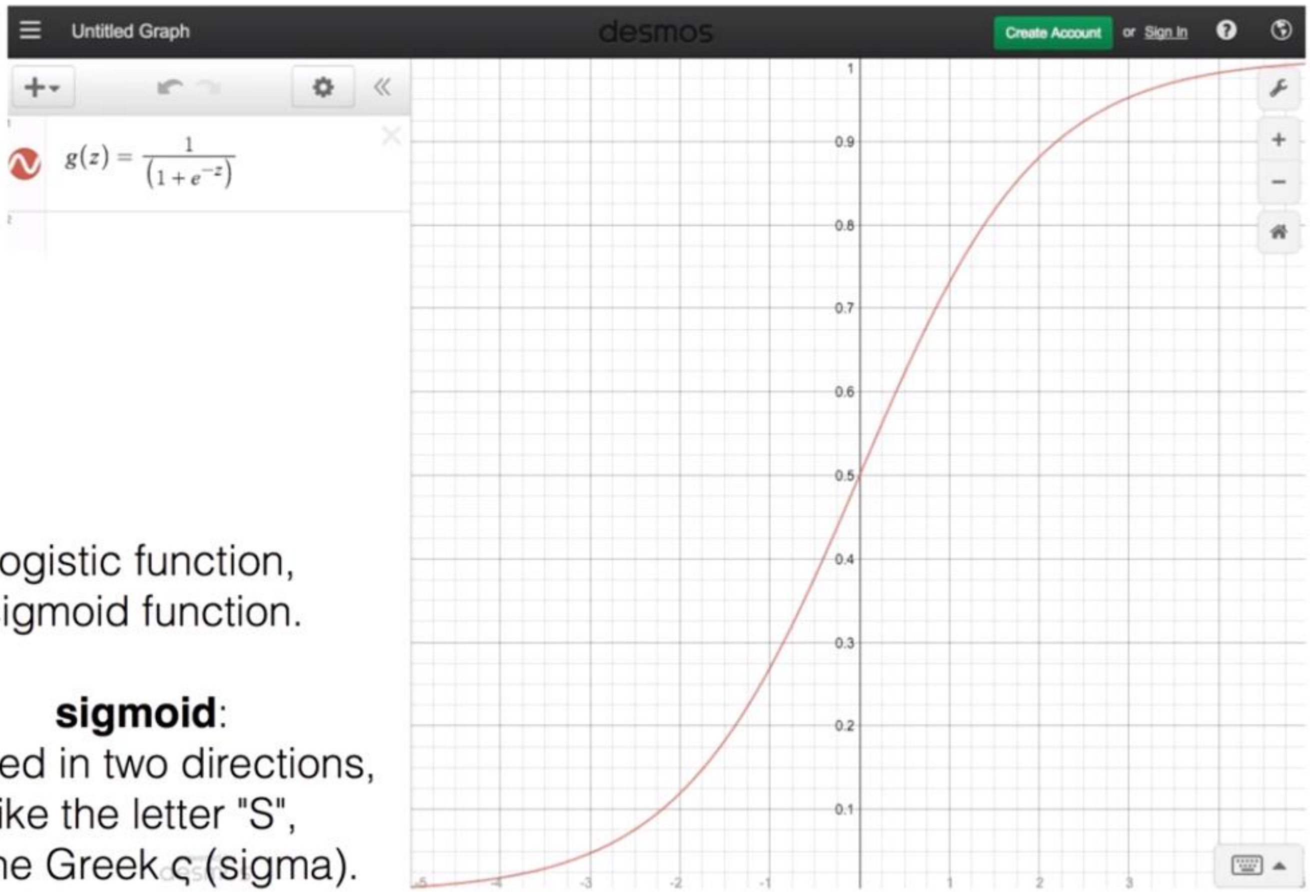
로지스틱 회귀 - Classification

Logistic Hypothesis

$$H(x) = Wx + b$$

로지스틱 회귀 - Classification

Sigmoid



logistic function,
sigmoid function.

sigmoid:
Curved in two directions,
like the letter "S",
or the Greek ς (sigma).

로지스틱 회귀 - Classification

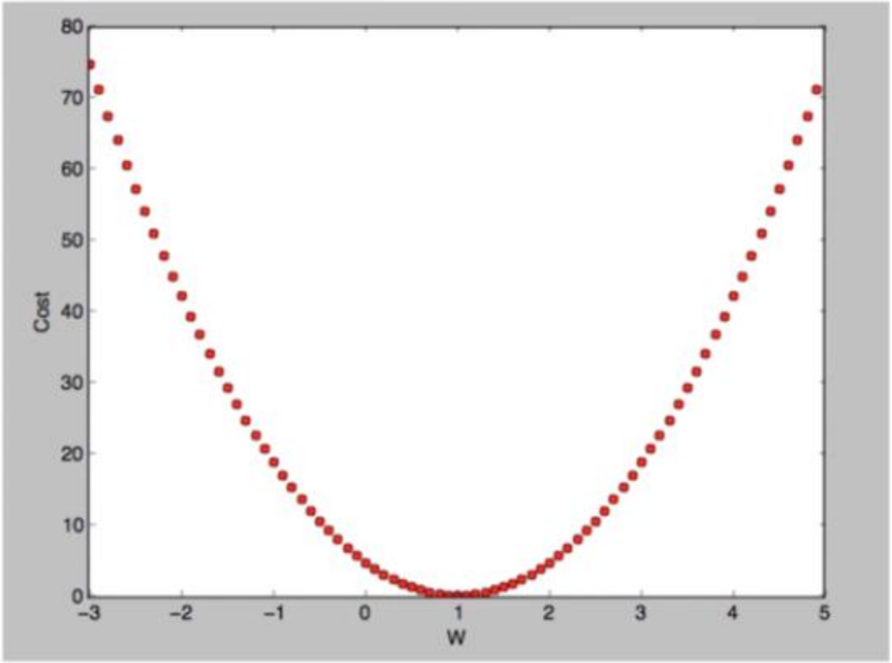
Logistic Hypothesis

$$H(X) = \frac{1}{1 + e^{-W^T X}}$$

로지스틱 회귀 - Classification

Cost

$$cost(W,b) = \frac{1}{m}\sum_{i=1}^{m}(H(x^{(i)}) - y^{(i)})^2 \quad \text{when} \quad H(x) = Wx + b$$



로지스틱 회귀 - Classification

Cost

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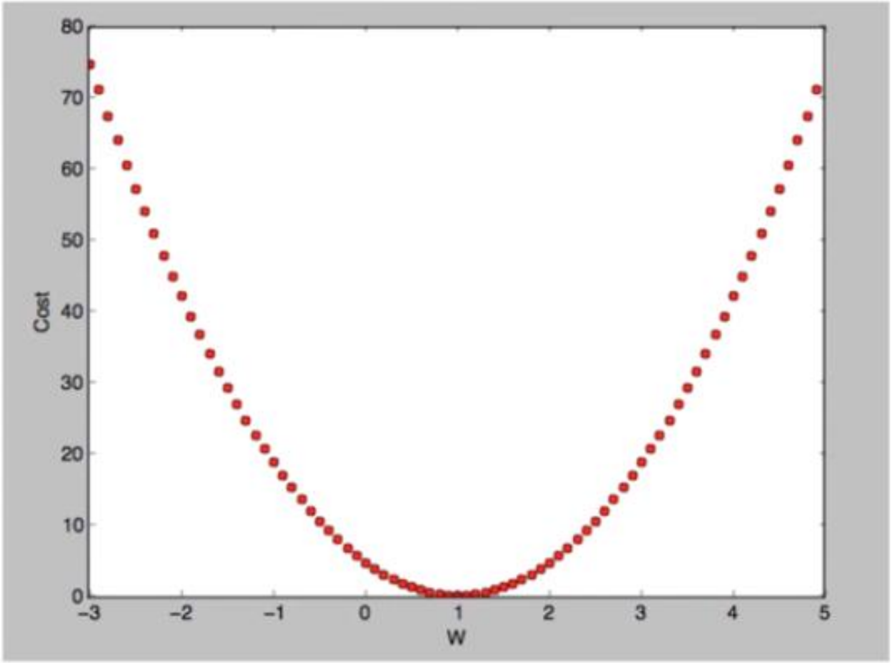


로지스틱 회귀 - Classification

Cost function

$$cost(W,b) = \frac{1}{m}\sum_{i=1}^{m}(H(x^{(i)}) - y^{(i)})^2$$

$$H(x) = Wx + b$$

$$H(X) = \frac{1}{1 + e^{-W^T X}}$$

로지스틱 회귀 - Classification

New cost function for logistic

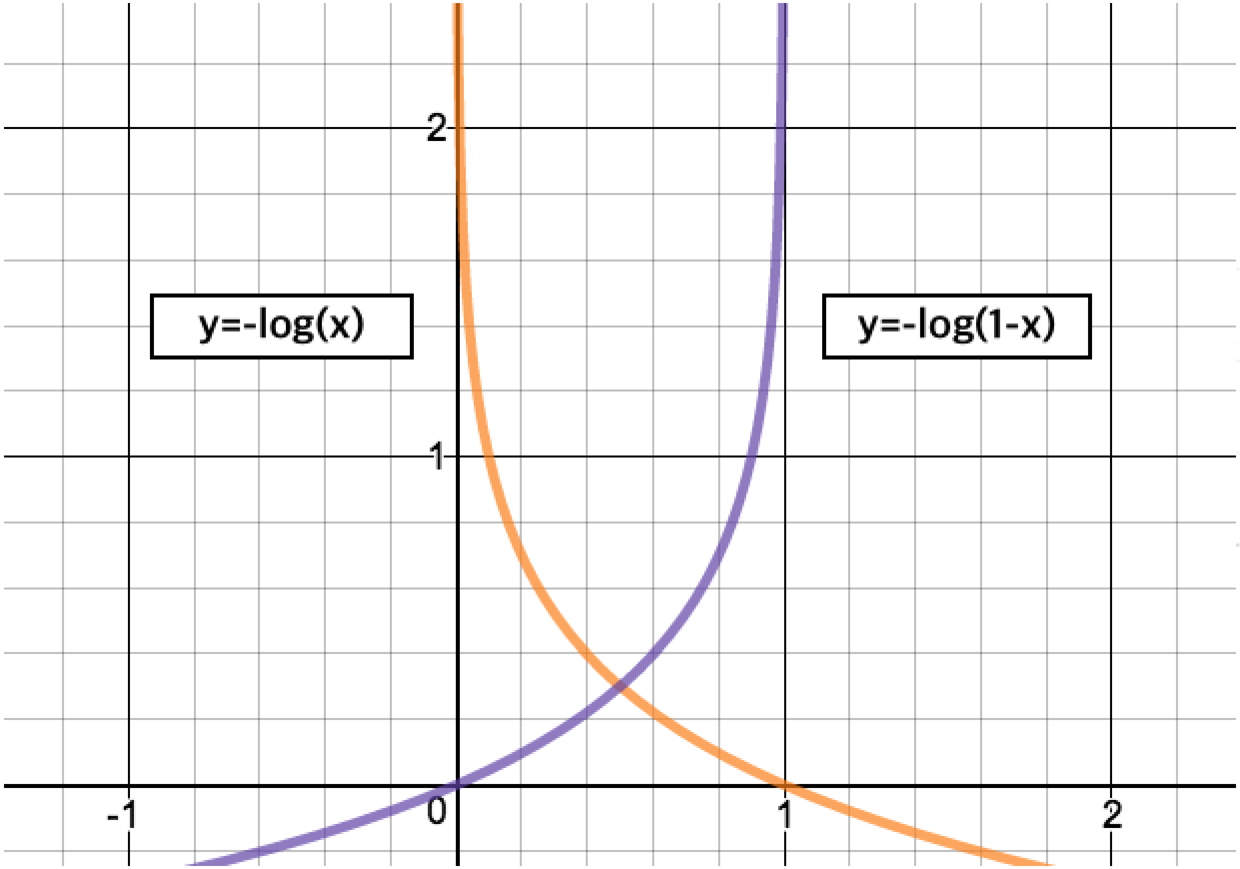
$$cost(W) = \frac{1}{m} \sum c(H(x), y)$$

$$c(H(x), y) = \begin{cases} -log(H(x)) & : y = 1 \\ -log(1 - H(x)) & : y = 0 \end{cases}$$

로지스틱 회귀 - Classification

understanding cost function

$$C(H(x), y) = \begin{cases} -log(H(x)) & : y = 1 \\ -log(1 - H(x)) & : y = 0 \end{cases}$$



로지스틱 회귀 - Classification

Cost function

$$cost(W) = \frac{1}{m}\sum c(H(x), y)$$

$$c(H(x), y) = \begin{cases} -log(H(x)) & : y = 1 \\ -log(1 - H(x)) & : y = 0 \end{cases}$$

$$c(H(x), y) = -ylog(H(x)) - (1 - y)log(1 - H(x))$$

로지스틱 회귀 - Classification

Minimize cost - Gradient decent algorithm

$$cost(W) = -\frac{1}{m}\sum ylog(H(x)) + (1-y)log(1-H(x))$$

$$W := W - \alpha\frac{\partial}{\partial W}cost(W)$$